# MOULI SAI SANTHOSH SOORAJ

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## EDUCATION

**SRM University-AP** **Expected 2027**

*Bachelor of Technology in Computer Science and Engineering* *Andhra Pradesh, India*

- Coursework: Data Structures & Algorithms, Computer Systems Architecture, Operating Systems, DBMS, Machine Learning.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, C++, Bash/POSIX Shell, SQL

**Web & Frameworks:** Next.js (App Router), React, Tailwind CSS, Node.js, Express, FastAPI, REST APIs

**Systems & DevOps:** Linux Kernel Basics, Git, Docker, Web Audio API, Microservices, CI/CD

**AI/ML & Tooling:** PyTorch, Scikit-Learn, Gemini API, OpenAI API, Local LLMs (Ollama), Edge Inference

## FEATURED PROJECTS

**Spotify Roaster • Next.js, Tailwind CSS, Spotify OAuth, Gemini AI** **Oct 2026**

- Architected fullstack web app analyzing user listening habits via Spotify OAuth token exchange with zero client secret exposure.
- Implemented concurrent API fetching using Promise.all across tracks, artists, and audio features, slashing data ingestion to <350ms.
- Built automated AI roast generation pipeline paired with dynamic Social Card preview generation using Next.js @vercel/og.
- Engineered local JSON cache persistence layer with rule-based fallback to guarantee 100% uptime during external API quota limits.

**Laya File Organizer • Python, Machine Learning, Systems, Local-First AI** **Oct 2026**

- Engineered a sub-35ms local file classifier and desktop organizer using non-autoregressive decision heuristics to sort downloads.
- Eliminated cloud API dependencies and network latency, achieving 100% offline data privacy and sub-second batch sorting throughput.
- Built robust cross-platform path resolution and safe-move file system semantics with rollback protection to prevent file corruption.

**Agent Setup CLI • Bash, POSIX Shell, Linux, DevOps, Systems Automation** **Oct 2026**

- Created a zero-dependency, idempotent developer environment bootstrap CLI cutting fresh machine setup from 4 hours to <40 seconds.
- Implemented non-destructive pre-flight environment probing that verifies existing toolchains before link creation.
- Engineered automated cross-platform compatibility layers supporting Ubuntu, Debian, macOS, and Arch Linux package managers.

**Quanevac • Python, Quantum Algorithms, Graph Optimization, Simulation** **Sep 2026**

- Researched and modeled evacuation pathfinding networks leveraging quantum heuristic state search and QAOA routing principles.
- Optimized classical simulation bottlenecks by bounding matrix dimensions and pruning exponential state-vector growth.

## LEADERSHIP & WRITING

**Open Source Contributor & Technical Writer** **2025 – Present**

- Author technical deep-dives on systems architecture, browser internals, and quantum computing on sooraj.fun and blog.sooraj.fun.
- Maintained open-source automation utilities and developer tools with over 20+ public repositories on GitHub.